

EDUCATION

Tianjin University

Tianjin, China

• B.S. in Data Science and Big Data Technology, School of Mathematics

Aug. 2023–Jul. 2027 (Expected)

◦ GPA: 3.41/4.0

◦ **Main Courses:** Mathematic Analysis , Probability Theory , Mathematical Statistics , Data Science , Real Analysis , Algorithm Design & Analysis , Ordinary Differential Equations (ODE)

RESEARCH INTERESTS

I aim to advance statistical learning theory to build reliable, interpretable models for complex data, focusing on:

AI for Science (AI4S)

• *Large-scale Data Analytics, Predictive Modeling, Deep Learning Applications.*

Biostatistics and Medical Statistics

• *DNA Data Storage, Neuroimaging Analysis, Survival Analysis.*

Statistical Learning & Inference

• *Optimization for Statistical Learning, Causal Inference, Semiparametric Modeling.*

RESEARCH EXPERIENCE

High-Dimensional Mediation Analysis in DNA Information Storage, Tianjin University

Tianjin, China

• Research Intern, advised by **Dr. Haixiang Zhang**

Sep. 2025–Present

- Proposed a deep partially linear model for high-dimensional survival mediation analysis with nonlinear confounding adjustment, enabling the identification of DNA sequence patterns governing storage failure.
- Achieved robust estimation: in simulations with 10,000 potential mediators, our method attained a 74.5% detection rate for key mediators and reduced estimation bias by up to 12.6% compared to conventional linear approaches.
- Manuscript in preparation for a targeted statistical/machine learning journal. (Anticipated submission: Summer 2026)

Forecasting Overdose ED Visits with LLM Agents, UNC-Chapel Hill

North Carolina, USA

• Research Intern, advised by **Dr. Tengfei Li**

Apr. 2026– Present

- Developed a forecasting system for nonfatal overdose Emergency Department(ED) visit rates across 51 US jurisdictions.
- Built a Claude-based LLM Agent that autonomously performs data analysis and prediction on unseen datasets without human intervention.

WORK EXPERIENCE

Teaching Assistant in Linear Algebra, Tianjin University

Tianjin, China

• Instructor **Prof. Yahui Shi**

Sep. 2025–Jan.2026

- Held weekly office hours for a class of 120+ students, clarifying concepts in linear Algebra
- Graded homework, projects, and mid-term exams; provided detailed feedback to improve student understanding.
- Assisted the professor in preparing lecture materials and updating the course syllabus.

HONORS & AWARDS

- **China Undergraduate Mathematical Contest in Modeling (CUMCM) - Second Prize (Provincial Level)** 2025
- **Mathematical Contest in Modeling (MCM) - Honorable Mention** 2025

SKILLS

Languages: Chinese Mandarin (Native), English (CET6 - 523)

Technical Skills: Proficient in coding: Python, $\LaTeX$, C++; Interested mathematical derivations; Solid foundation in mathematics and statistics.

Teamwork: Technical communication, Collaborative research.